

Time to prioritise community health workers: a decade of cost-effectiveness evidence



Governments in low-income and middle-income countries face a stark challenge: delivering quality health care within the shrinking fiscal space. Many governments default to facility-based, curative investments, yet this model alone cannot deliver universal health coverage. Hospitals often struggle to reach remote communities, deliver preventive care, and address social and behavioural drivers of disease—areas in which community health workers (CHWs) excel. When CHWs are adequately trained, supervised, supported, and compensated, they can improve health outcomes, lower costs, reduce inequities, and strengthen health systems.¹ The economic case is equally compelling: returns are worth up to ten times the investment in terms of improved health outcomes and savings from averted health-care costs.² Well designed economic evaluations further equip policy makers with the granular, context-specific evidence they need—including the cost of inaction—to allocate resources and maximise health gains.³

The Community Health Impact Coalition's series of five scoping reviews, spanning 130 studies and 380 scenarios, provides evidence that CHWs are a cost-effective strategy to expand essential services and reinforce health systems. Building on the 2015 review by Vaughan and colleagues,⁴ these reviews summarise economic evidence published since 2015, spanning five major service areas: horizontal (ie, addressing more than one disease) programmes;⁵ HIV, tuberculosis, and malaria;⁶ non-communicable diseases (eg, hypertension, diabetes, and human papillomavirus infection or cervical cancer) and mental health (O'Donovan J and colleagues, unpublished); reproductive, maternal, newborn, and child health (Miyares M and colleagues, unpublished); and neglected tropical diseases (Stansert Katzen L and colleagues, unpublished). We included both partial and full economic evaluations and reported the authors' conclusions on the cost-effectiveness (ie, the comparison of costs and health outcomes, relative to defined thresholds or alternative services) and affordability of CHW programmes (figure).

CHW programmes consistently delivered strong health outcomes more cost-effectively than facility-based care or other modalities, making them a sound investment, especially in resource-limited settings.^{2,5,6} Across reviews,

the median annual cost per capita, standardised to 2024 US dollars, was just US\$0.59, and \$10.03 per beneficiary. Unit costs varied widely; for example, per capita costs ranged from \$0.29 to \$67.95 for horizontal interventions and from \$0.23 to \$1.33 for non-communicable disease interventions. Costs per beneficiary ranged from \$1.20 to \$26 556 for HIV, tuberculosis, and malaria services and from \$0.02 to \$1547 for reproductive, maternal, newborn, and child health.

The review covered 380 scenarios, including 255 cost-effectiveness analyses. CHW programmes were cost-effective in 78–93% of scenarios for reproductive, maternal, newborn, and child health; 81–85% of scenarios for non-communicable diseases and mental health; and 83–88% of scenarios for HIV, tuberculosis, and malaria.

Lancet Prim Care 2025; 1: 100076

Published Online November 29, 2025
<https://doi.org/10.1016/j.lanprc.2025.100076>

For the **scoping reviews** see <https://joinchic.org/resources/cost-effectiveness/>

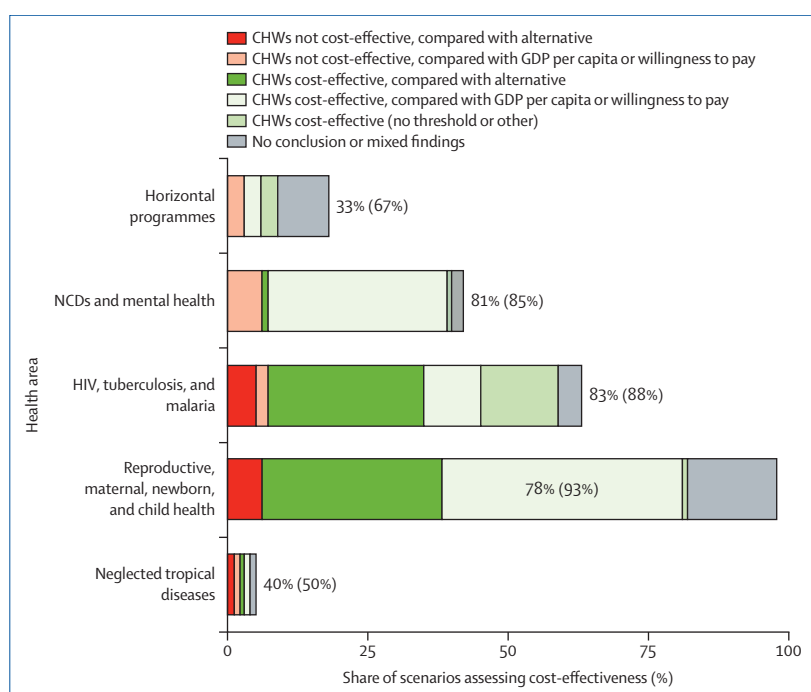


Figure: Conclusions on cost-effectiveness in the 380 scenarios of Community Health Impact Coalition's series of five scoping reviews, by health area

Scenarios represent unique combinations of parameters such as country, intervention model, or study perspective. One study could contribute multiple scenarios. Percentages show the share of scenarios deemed cost-effective. Percentages in parentheses exclude scenarios with unclear conclusions. Each review defined cost-effectiveness in the same way. Cost-effectiveness judgements were made by authors in comparison with alternative models of care or other thresholds. GDP-per-capita thresholds are criticised for ignoring local opportunity costs and affordability.⁷ We excluded comparisons between CHW programmes to assess overall cost-effectiveness of CHW programmes. CHWs=community health workers. GDP=gross domestic product. NCDs=non-communicable diseases.

Not all CHW programmes were equally cost-effective, and comparative studies show that programme design matters.⁸ Our review found that integrated approaches, such as linking CHWs to primary care facilities, using digital tools, formalising employment, or running multidisease campaigns, tend to be more cost-effective than stand-alone models that are not integrated into the primary health care system and are focused on one disease only. For example, integrated CHWs in Liberia cut diagnostic and treatment costs for neglected tropical diseases by up to ten times, compared with single-disease approaches.⁹

CHW-led interventions for HIV, tuberculosis, and malaria often outperformed facility-based models in both cost-effectiveness and health outcomes, particularly for the most susceptible populations. For example, in Kenya and Tanzania, CHWs providing HIV services to people who were pregnant and individuals with no co-infections had better outcomes at lower costs than models involving more highly trained health workers.¹⁰ This finding emphasises the ability of CHWs to deliver effective, equitable care to remote populations.

Cost-effectiveness does not guarantee affordability, yet few studies assess this aspect. For affordability, we noted whether authors reported on budget impact, including whether the intervention was affordable within current budget constraints. Of the 380 scenarios reviewed, only 89 examined affordability; 34 (38%) of 89 deemed the intervention affordable, eight (9%) deemed it unaffordable, and 47 (53%) were inconclusive. This evidence gap matters. Unit costs for CHW programmes vary widely. For example, in Liberia, where CHWs handle more than half of malaria cases, dedicating 25% of the health budget to community health could be justified. Yet, in settings with different fiscal or epidemiological realities, even a 6-7% allocation could raise questions about affordability.⁸

Importantly, the true cost-effectiveness of CHWs could be underestimated because they also address and improve key social determinants of health. With strong evidence available since 2015, policy makers and funders should prioritise CHWs as a cost-effective foundation of primary health care and a cornerstone of achieving universal health coverage.

LSK and CB are joint first authors. KV is an independent consultant in the field of global public health and health economics and receives consulting fees from War Child, Columbia University, UNICEF, Bitran & Asociados, Genesis Analytics, GFA Consulting, Panorama Global, the Impaction Foundation, Results for Development, Monash University, the Royal Tropical Institute, Cerebral Palsy Africa, Tunafasi–Appui au Développement de l'Enfant en Détresse, and Population

Services International. DP receives consulting fees from WHO and Vanna Health; receives honoraria from Harvard University for participation in an executive leadership primary health care training course; receives financial support for travel for participation in events by the Community Health Impact Coalition (CHIC) and an event by the Fleming Initiative; is a co-founding board member of the CHIC and the Financing Alliance for Health, unpaid; and is a board member of the Hyde Square Task Force, unpaid. All other authors declare no competing interests.

Copyright © 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

*Linnea Stansert Katzen, Cleo Baskin, Kelsey Vaughan, James O'Donovan, Madelyn Miyares, Carey Westgate, Josef Ernst, Dickson Mbeve, Harriet Napier, Zeus Aranda, Daniel Palazuelos, Marius Nkenfack, Soleine Scotney, Shivani Shukla, Madeleine Ballard
linnea.stansert-katzen@joinchic.org

Community Health Impact Coalition, Seattle, WA 98121, USA (LSK, CB, JO, CW, JE, MB); Institute for Life Course Health Research, Department of Global Health, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa (LSK); Centre for Health and Sustainability, Department of Women's and Children's Health, Uppsala University, Uppsala, Sweden (LSK); ResiliSUS Lab, Fundação Oswaldo Cruz, Rio de Janeiro, Brazil (CB); Bang for Buck Consulting, Amsterdam, Netherlands (KV); Arnhold Institute for Global Health, Icahn School of Medicine at Mount Sinai, New York City, NY, USA (JO, MB); Department of Health Policy and Management, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA (MM); Kasungu District Hospital, Kasungu, Malawi (DM); Clinton Health Access Initiative, Boston, MA, USA (HN); Compañeros En Salud, Ángel Albino Corzo, Mexico (ZA); Partners in Health, Boston, MA, USA (DP); Department of Global Health and Social Medicine and Division of Global Health Equity, Brigham and Women's Hospital, Medical School, Harvard University, Boston, MA, USA (DP); Division of Global Health Equity, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA (DP); Muso, Bamako, Mali (MN); Financing Alliance for Health, Nairobi, Kenya (SSC, SSH)

- 1 WHO. What do we know about community health workers? A systematic review of existing reviews. 2021. <https://www.who.int/publications/i/item/what-do-we-know-about-community-health-workers-a-systematic-review-of-existing-reviews> (accessed Aug 10, 2025).
- 2 Dahn B, Woldemariam AT, Perry H, et al. Strengthening primary health care through community health workers: investment case and financing recommendations. July, 2015. <https://chwccentral.org/wp-content/uploads/2015/09/CHW-Financing-FINAL-July-15-2015.pdf> (accessed Aug 15, 2025).
- 3 Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. Methods for the economic evaluation of health care programmes. Fourth edition. Oxford University Press, 2015.
- 4 Vaughan K, Kok MC, Witter S, Dieleman M. Costs and cost-effectiveness of community health workers: evidence from a literature review. *Hum Resour Health* 2015; **13**: 71.
- 5 O'Donovan J, Kumar MB, Ballard M, et al. Costs and cost-effectiveness of integrated horizontal community health worker programmes in low- and middle-income countries (2015–2024): a scoping literature review. *BMJ Glob Health* 2025; **10**: e017852.
- 6 O'Donovan J, Baskin C, Stansert Katzen L, et al. Costs and cost-effectiveness of community health worker programs focussed on HIV, TB and malaria infectious diseases in low- and middle-income countries (2015–2024): a scoping literature review. *PLoS Glob Public Health* 2025; **5**: e0004596.
- 7 Bertram MY, Lauer JA, De Joncheere K, et al. Cost-effectiveness thresholds: pros and cons. *Bull World Health Organ* 2016; **94**: 925–30.
- 8 Taylor C, Griffiths F, Lilford R. Affordability of comprehensive community health worker programmes in rural sub-Saharan Africa. *BMJ Glob Health* 2017; **2**: e000391.
- 9 Godwin-Akpan TG, Diaconu K, Edmiston M, et al. Assessing the cost-effectiveness of integrated case management of neglected tropical diseases in Liberia. *BMC Health Serv Res* 2023; **23**: 705.
- 10 Cherutich P, Farquhar C, Wamuti B, et al. HIV partner services in Kenya: a cost and budget impact analysis study. *BMC Health Serv Res* 2018; **18**: 721.

For examples of different fiscal or epidemiological realities see <https://www.exemplars.health/>